Balancing

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Sampling – ML: Why?

- **Class imbalance is prevalent in many applications:** fraud/intrusion detection, risk management, text classification, medical diagnosis/monitoring, etc.

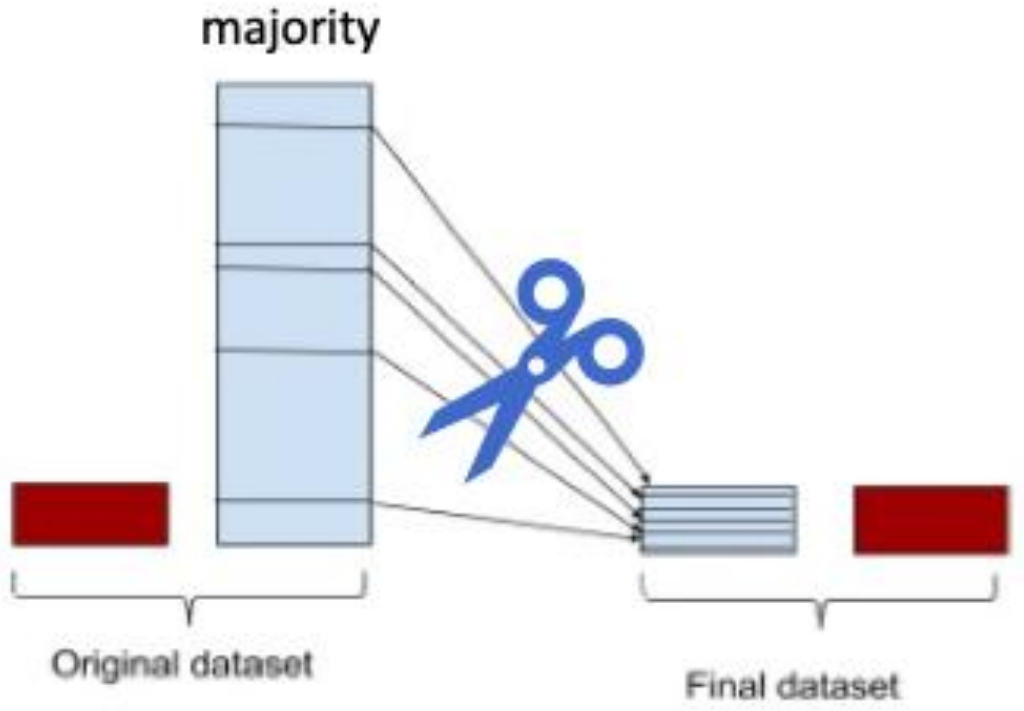
- **Standard classifiers tend to be overwhelmed by the large classes and ignore the small ones,** i.e., tend to produce high predictive accuracy over the majority class, but poor predictive accuracy over the minority class

- **A balanced input might help the classifiers.**

Easy example

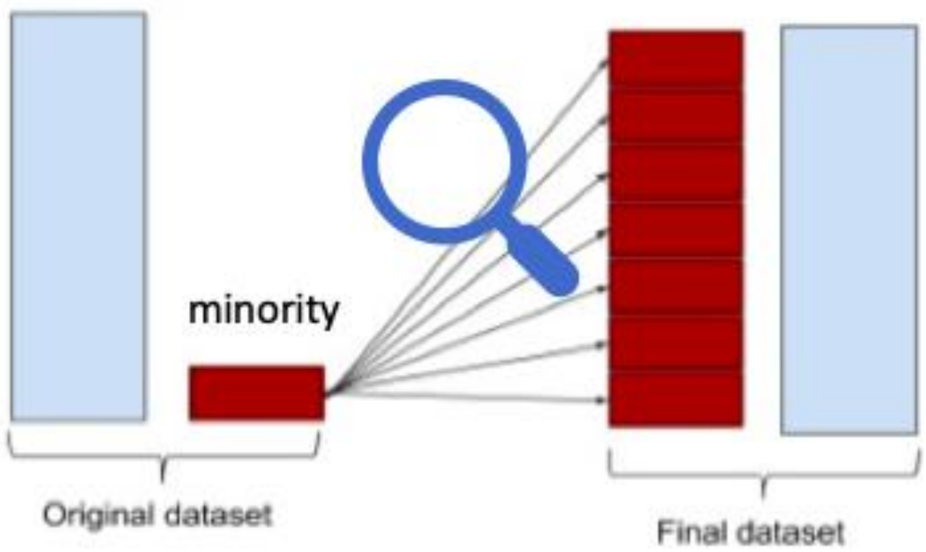
ML Types – Undersampling

- Extract a smaller set of majority instances while preserving all the minority instances



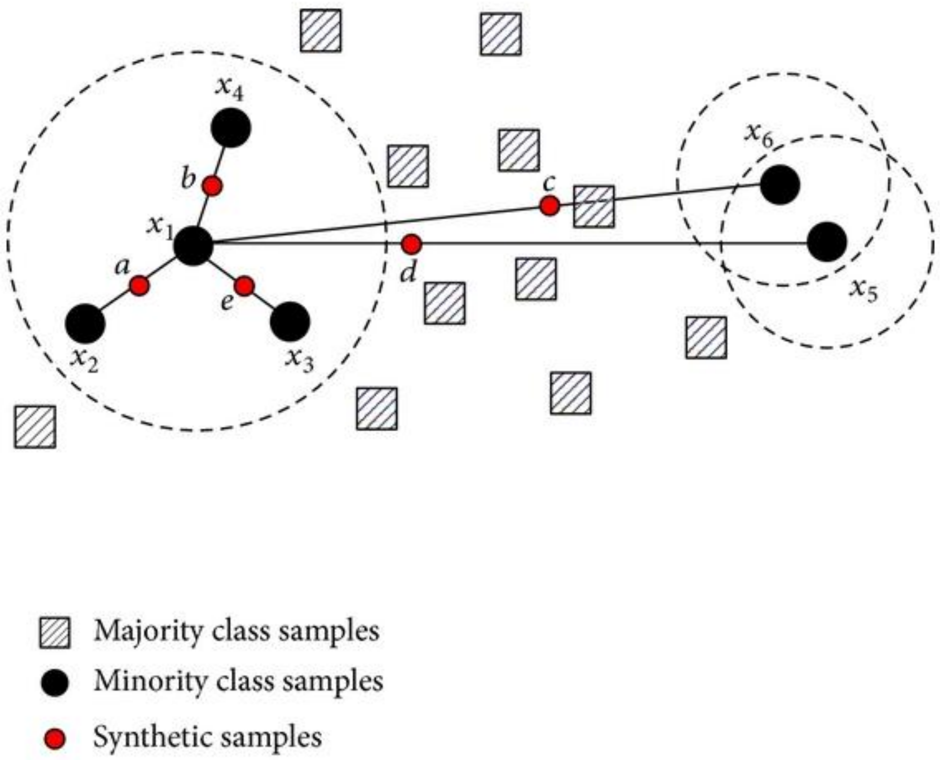
ML Types – Oversampling

- Increases the number of minority instances by over-sampling them.

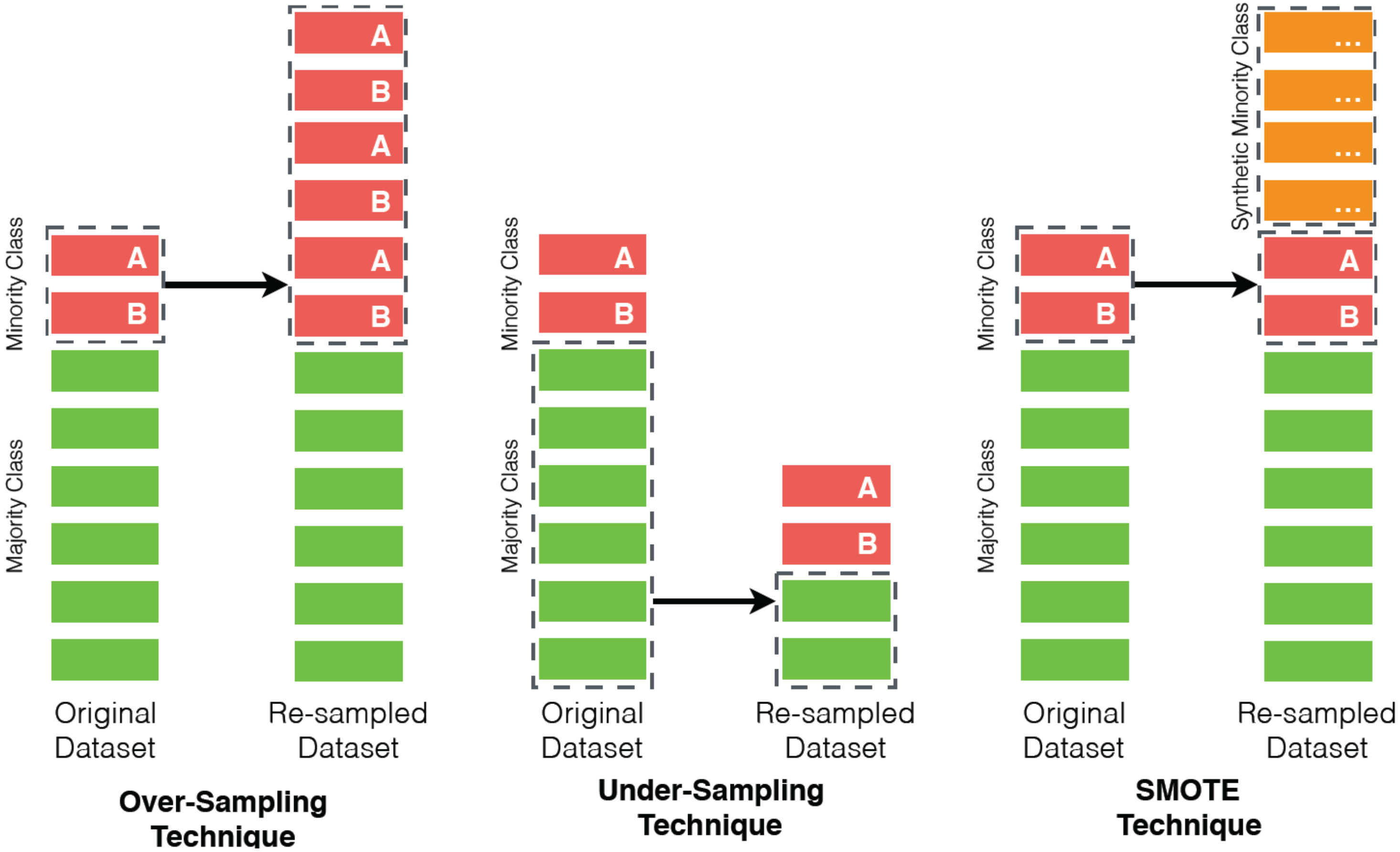


ML Types: Synthetic Minority Oversampling Technique (SMOTE)

- Generates synthetic data based on the feature space similarities between existing minority instances.



SUMMARY



Minority Class
Majority Class
A
B
A
B
A
B
A
B
Original Dataset
Re-sampled Dataset
Over-Sampling Technique
Minority Class
Majority Class
A
B
A
B
Original Dataset
Re-sampled Dataset
Under-Sampling Technique
Minority Class
Majority Class
Synthetic Minority Class
A
B
...
...
...
...
A
B
Original Dataset
Re-sampled Dataset
SMOTE Technique

Lab

- Compare the accuracy with/without oversampling.
  - Take one dataset.
  - Take three classifiers: RandomForest / NaiveBayes / Ibk
  - Perform 66/33 ordered split.

WEKA API

- Undersapling
  - weka.filters.supervised.instance.**SpreadSubsample** -M 1.0

- Oversampling
  - noReplacement=false, biasToUniformClass=1.0, and sampleSizePercent=Y, Y = 100 * (majority –minority)/minority.
  - Example for the diabetes data: weka.filters.supervised.instance.Resample -B 1.0 -Z 130.3
- SMOTE
  - If needed, download it at https://weka.sourceforge.io/packageMetaData/SMOTE/index.html